

Proof WS 2

- Use a direct proof to prove each of the following statements.
 - The product of two odd integers is odd.
 - The sum of two consecutive odd numbers is divisible by 4.
- Use a contrapositive proof to prove each of the following statement: If a and b are integers and ab is even, then at least one of a and b is even.
- Use a proof by contradiction to prove each of the following statement: $\sqrt{5}$ is irrational.
- Prove each of the following logical equivalence: Let n be a positive integer. $n - 3$ is odd if and only if $n + 2$ is even.
- Prove that the number $1 - 5\sqrt{2}$ is irrational.
- It is known that $\sqrt{n}$ is irrational whenever n is a positive integer that is not a perfect square. Use this result to help prove that $\sqrt{6} + \sqrt{10}$ is irrational.
- Write each sentence in mathematical notation.
 - For all positive real numbers x and y , if x is greater than y , then the square of x is greater than the square of y .
 - There exists a rational number c between integers a and b where $a < b$, such that c is the average of a and b .
 - Let n be a positive integer such that for all n , $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$.
- Prove by contradiction that $\log_5 13$ is an irrational number.
- It is given that $a > 0, b > 0$ are real numbers. Consider the statement $\forall a \left(\forall b, \log_{\frac{1}{a}} \frac{1}{b} = \log_a b \right)$. Prove that the statement is true.
- Use proof by contradiction to show that $\log_2 5$ is irrational.
- It is given that p and q are both real numbers. Consider the statement $\forall p \left(\forall q, \frac{1}{p^2} < \frac{1}{q^2} \right)$. Either prove that the statement is true or provide a counterexample.

Answer:

7. a. $\forall x, y \in \mathbb{R}, x, y > 0, (x > y) \Rightarrow (x^2 > y^2)$; b. If $a, b \in \mathbb{Z}, a < b, \exists c \in \mathbb{Q}, a < c < b: c = \frac{a+b}{2}$; c. Let $n \in \mathbb{Z}, n > 0: \forall n, 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$.